Order statistics exercises

09-E01 Comparisons for both extremes

Difficulty: Foundational.

Find minimum and maximum together by pairing for n=11. Count comparisons and verify the general formula.

09-E02 Random-select interval conversion

Difficulty: Advanced.

Among ten elements seek rank k=7. A pivot leaves four smaller-side elements, itself, and five greater-side elements. Which side and local rank come next? What changes for k=3?

09-E03 Median-of-medians exclusion

Difficulty: Research.

Group n elements into fives and select group medians. With three-way partition and duplicate keys allowed, prove at least $3n/10-6$ values are $\leq$pivot and at least as many $\geq$pivot. Explain removal of the short and pivot groups. Give the conservative bound for n=50.

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09-E04 Local ranks with duplicates and a nonzero start

Difficulty: Advanced.

A[4..10] contains [8,3,5,3,9,1,3], with pivot 3. Three-way partition gives lt=5,gt=7. Determine the next action for local ranks 1,4,6. Why can k not be compared directly with lt?

09-E05 Median-of-medians is not the exact median

Difficulty: Research.

Five sorted groups are [1,2,4,17,25], [3,5,8,18,24], [6,9,12,19,23], [7,10,14,20,22], [11,13,15,16,21]. Give the pivot, region sizes, and next local rank when seeking the sixteenth smallest. Compare with the actual overall median.

09-E06 Average larger side and expectation induction

Difficulty: Research.

Order statistics exercises (continued)

For n=6 list max(r−1,6−r) for every distinct pivot rank r and average them. With nonrecursive cost at most cn and shorter-state expectation at most a times length, derive a linear upper bound. Why can average size not simply be inserted into an arbitrary T?

Work independently before opening the separate solutions PDF.

References and attribution

Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein. Introduction to Algorithms, 3rd Edition.

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